Imitating a Coding Agent at Low Cost: Offline Distillation of Fable 5 Traces into Qwen3-8B

Deep Gandhi
Transformer Lab
deep@lab.cloud

Abstract

Large agentic models such as Fable 5 reason and call tools to solve software tasks, but they are expensive to run. We ask whether a small open model can absorb that behavior from a *static dump* of the larger model's execution traces, with no teacher logits and no live environment. Using 4,665 Fable 5 coding-agent steps (60 sessions), we apply span-masked supervised fine-tuning, predicting the teacher's reasoning and next action from the running transcript, to Qwen3-8B with a small LoRA adapter. On a group-disjoint held-out test split, the distilled student cuts next-step perplexity by 50% (negative log-likelihood $1.61 \to 0.93$), with the largest gain on the free-text reasoning the base model handled worst. The cost to general ability is small: HumanEval pass@1 and GSM8K accuracy are unchanged ($0.87 \to 0.86$ and $0.92 \to 0.92$), while MBPP pass@1 falls modestly ($0.85 \to 0.79$). A controlled sweep shows the recipe is simple and robust: two epochs on the full, unfiltered data beats aggressive LLM-judge filtering, SVD-based (PiSSA) initialization, larger LoRA rank, and a third epoch, all of which hurt. We release the training and evaluation code, the adapter, and a full reproducibility package. We do *not* report an end-to-end agentic-coding resolve rate (e.g. SWE-bench): single-shot patch generation from an 8B model rarely yields applying patches, so a multi-step agentic harness is the appropriate evaluation, which we leave as the main open question.

1 Introduction

Frontier coding agents interleave chain-of-thought reasoning with tool calls (read a file, run a shell command, edit code) inside a multi-turn loop. Their capability is valuable but their cost and closed weights motivate *distilling* that behavior into a small, open student. The usual obstacles are sharp here: the teacher is closed (no logits for KL distillation), the data is a fixed offline dump (no on-policy correction, no environment to query), and the corpus is tiny and narrow (60 sessions from one harness). Prior work warns that offline behavioral cloning of multi-turn agents suffers covariate shift [1, 2] and that supervised fine-tuning can *degrade* a small model's base capability unless carefully controlled [3, 4].

We study the most constrained version of the problem: *offline, sequence-level* distillation of Fable 5 (`claude-fable-5`) agentic coding traces into Qwen3-8B, and ask two questions. (1) Does it work: does the student imitate the teacher's next step better than the untouched base? (2) What does it cost: does specializing on coding traces erode general code and math ability?

1. We show offline span-masked SFT distills agentic coding behavior effectively: a 50% cut in held-out next-step perplexity over the base (§5), with the biggest gain on free-text reasoning (§5, subgroup analysis).

2. We show the gain comes at a small cost to general ability: HumanEval and GSM8K are unchanged, with one modest regression on MBPP (0.85→0.79) (§5).

3. Through a controlled sweep we show the effective recipe is *simple*: full unfiltered data and two epochs beat LLM-judge filtering, PiSSA initialization, higher LoRA rank, longer training, and a lower learning rate (§5, ablations). On clean data, curation removes signal.

2 Related Work

Reasoning and agent distillation. Distilling chain-of-thought into small students is well studied, often via teacher logits or multi-teacher merging [5, 6]. Kajitsuka et al. [3] caution that hard CoT distillation into small Qwen models frequently *degrades* the base unless gated against a pre-distillation baseline, a warning we adopt as a success criterion. Closest to us, Liu et al. [7] distill a ReAct tool-use agent by decoupling reasoning and action spans, and Du et al. [8] distill tool-use reasoning into a ∼7B student; both inform our span-masked objective, though their mechanisms assume teacher logits we lack.

Training agents from trajectories. Recent systems fine-tune software-engineering agents on teacher trajectories [9, 10]. Jiang et al. [9] report that 239 curated samples can beat 66k and that *unfiltered* traces cause catastrophic unlearning, motivating our filtered-vs-full ablation. Raju et al. [10] analyze SWE-bench as a benchmark and the harness needed to run it.

The offline-imitation problem. Lauffer et al. [1] show offline behavioral cloning of multi-turn LM agents suffers covariate shift; Li et al. [2] adapt DAgger to LLM agents but require online teacher queries, and Li and Zhang [11], Wulfmeier et al. [12] give theory and IRL alternatives that also need interaction. With a static dump we cannot apply these fixes, so we lean on data quality and regularization instead.

Data curation and PEFT. Quality filtering and rejection sampling can help when data is noisy [13, 14]; Iskander et al. [13] show label-free LLM-judge filtering helps tool-use SFT. We test this directly and find it *hurts* on our already-clean corpus. For parameter-efficient tuning we use LoRA and compare against PiSSA-style SVD initialization [15].

3 Method

Setup. The teacher signal is a static set of Fable 5 agent steps. Each step is a tuple of a running *context* (the user task plus prior transcript), the teacher's *reasoning* trace, and its next *output* (a tool call or assistant text). We frame distillation as next-step imitation: given the context, the student must produce the reasoning followed by the next action.

Objective. We do offline sequence-level SFT with a *span-masked* loss. For a step with context c and target t (the teacher's <think>reasoning</think> plus serialized action), the loss is the token-level negative log-likelihood over t only:

$$\mathcal{L} = -\sum_i \log p_\theta(t_i \mid c,\, t_{<i}), \tag{1}$$

with context tokens and tool *observations* masked out. There is no logit/KL term: the teacher is a closed model and the dump contains only text. We tune a LoRA adapter on the attention and MLP projections, leaving base weights frozen. This is deliberately gentle: a low-rank, few-epoch update we hypothesize will specialize the student without overwriting general ability.

4 Experimental Setup

Data. `Glint-Research/Fable-5-traces` (`fable5_cot_merged.jsonl`, sha256 `49dbd56b...`, AGPL-3.0): 4,665 steps from 60 unique sessions; 81% are tool calls (Bash, Edit, Read, Write, ...), 19% assistant text. There are no final-answer or correctness labels. We split *group-aware by session* (so no session straddles splits) into 80/10/10 by rows: train 3,732 (30 sessions), val 467 (16), test 466 (14). Contexts are capped at 7,022 characters by the source.

Students and training. Qwen3-8B (primary) and Qwen2.5-Coder-7B (comparison). LoRA r=32, α=32, dropout 0.05; learning rate 2×10^{-4} cosine, batch 1 with gradient accumulation 8, bf16, gradient checkpointing, seed 42. The final model is two epochs on the full train split, and trains in about two GPU-hours on a single 40 GB A100, so the result is cheap to reproduce.

Evaluation. The primary metric is teacher-forced next-step *negative log-likelihood* (NLL) on the held-out *test* split, reported overall and per output type. We probe transfer with three external benchmarks generated with Qwen3 "thinking" disabled for bounded, parseable output: HumanEval and MBPP pass@1 (code, by subprocess execution) and GSM8K exact-match (reasoning), n=100 each. We do not run end-to-end agentic benchmarks (see Limitations).

5 Results

Distillation works, at a small cost. Table 1 and Figure 1 report the headline. On the held-out test split the distilled student cuts NLL from 1.6120 to 0.9254 (−0.687; perplexity 5.01 → 2.52, −50%). General ability is largely preserved: HumanEval pass@1 (n=100) moves 0.87 → 0.86 and GSM8K (n=100) is flat at 0.92, both within sampling noise at this n. The one real cost is on MBPP (n=100), which falls 0.85 → 0.79. The model meets both required success criteria (beat base on the held-out trace evaluation; do not regress HumanEval code generation), with the MBPP drop noted as a genuine, if small, specialization cost.

Where the gain lands. The teacher's steps split into tool calls (81%) and free text (19%). The base model is far worse at free text (NLL 2.03) than at tool calls (1.53). Distillation improves *both* but helps free text most (−0.776 vs. −0.669), narrowing the gap from 0.50 to 0.40 nats (Figure 1). The student absorbs the teacher's explanatory reasoning exactly where the base was weakest.

Table 1: Headline results vs. target gates. Held-out *test* split for NLL; n=100 for the external benchmarks. Lower is better for NLL/perplexity.

Metric	Qwen3-8B base	fabledistil	Target gate
Test NLL ↓	1.6120	**0.9254**	beat base ✓
Test perplexity ↓	5.01	**2.52**	(−50%)
HumanEval pass@1	0.87	0.86	no regression ✓
MBPP pass@1	0.85	0.79	small regression
GSM8K (exact match)	0.92	0.92	report (flat)

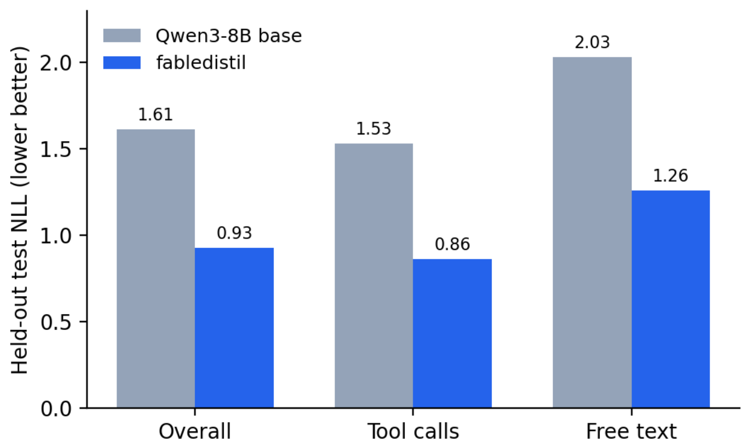

Figure 1: Held-out test NLL, overall and by output type. Distillation improves every subgroup and most on free text, where the base was weakest.

The effective recipe is simple. A controlled sweep on Qwen3-8B (Figure 2, validation NLL) shows the winning configuration is unremarkable and robust. Two epochs (0.9091) beat one (0.9228); a third epoch *overfits* the 60-session corpus (0.9313). LoRA rank is a non-lever ($r \in \{16, 32, 64\}$ all 0.9089–0.9097). Three intuitively "smarter" choices all *hurt*: aggressive LLM-judge filtering to the top 60% (0.9311), PiSSA initialization (0.9378), and a lower learning rate 1×10^{-4} (0.9210). The filtering result is the sharpest: because the corpus is already clean (near-zero degenerate traces), curation removes signal rather than noise, the opposite of the noisy-data regime where filtering helps [13, 9]. The effect is not model-specific: Qwen2.5-Coder-7B also improves under the same recipe (base 1.4297 → 0.9358), at a similar small code cost (HumanEval 0.90→0.86, MBPP 0.82→0.76).

6 Discussion

The distilled student passes its gates with a large margin and only a mild specialization cost. Prior work warns that fine-tuning small models on specialized data can sharply degrade base capability [3, 4]; here the effect is limited to a single benchmark (MBPP, −0.06), with HumanEval and GSM8K unchanged. We attribute the mildness to the gentleness of the update: a low-rank adapter, two epochs, and clean full data form a broad, stable basin rather than a fragile peak, and

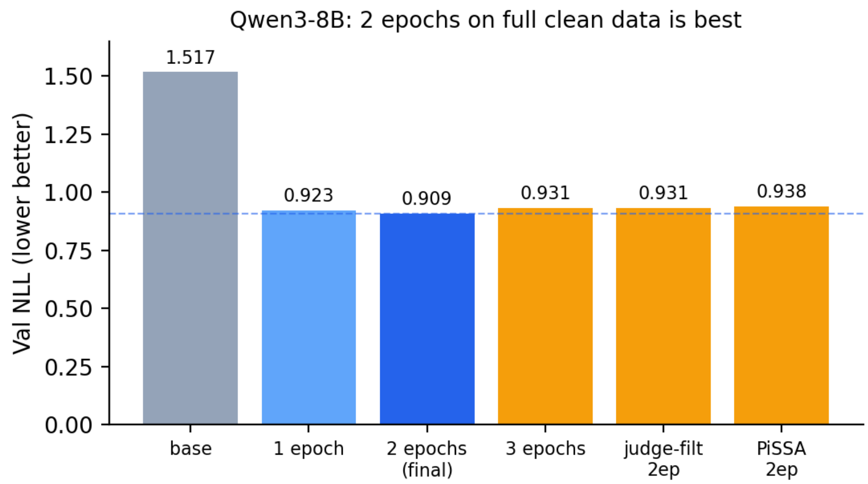

Figure 2: Qwen3-8B validation NLL across the sweep. Full, unfiltered data with two epochs (dashed line) is best; longer training, filtering, and PiSSA all regress.

every neighboring configuration we tried is close or worse, not better. The subgroup result suggests the teacher's distinctive value, and what the student most needs, is the free-text reasoning *between* actions, not the tool syntax it already largely knows.

7 Limitations

Construct validity: single-step, not agentic. Our primary metric is teacher-forced next-step NLL; it does not measure end-to-end task success, where compounding error under covariate shift, the very failure mode flagged for offline behavioral cloning [1, 2], could erode the single-step gains. We attempted an end-to-end SWE-bench Lite evaluation under a single-shot setting (the model is given the issue and relevant files and must emit a patch, graded by executing the repository tests). The pipeline runs end to end, but the 8B student's single-shot patches rarely apply cleanly, yielding no resolved instances on a small sample. A multi-step agentic harness, in which the model reads files and makes verified edits, is the appropriate evaluation and the setting our distillation most directly targets; we leave it as the main open question. **External validity:** 60 sessions from one agent harness; long-tail tools are under-represented and broad generalization is unproven. **Internal validity:** source contexts are truncated at 7,022 characters, bounding achievable loss on long sessions; benchmark transfer is measured with thinking disabled and n=100, so small benchmark differences are within sampling noise. The stretch goal of matching a larger same-family model was not tested.

8 Availability

All of our artifacts, the training and evaluation code, the group-aware data-split script, the LoRA adapter, and the reproducibility package (pinned environment, seed, data hash, exact commands), are available from the authors on request only. The training data, `Glint-Research/Fable-5-traces`, is public under AGPL-3.0; the distilled adapter inherits that provenance. Base weights (Qwen3-8B) are under their own license.

9 Conclusion and Future Work

Offline, logit-free distillation of a frontier coding agent's traces into a small open model works and is cheap: a two-epoch LoRA on 3,732 steps halves held-out next-step perplexity on Qwen3-8B at a small cost to general ability (one modest MBPP regression, HumanEval and GSM8K unchanged), and the best recipe is the simplest one. The clear next step is end-to-end agentic evaluation (SWE-bench with a multi-step harness) and, if covariate shift bites, a budget-friendly pseudo-on-policy round (student rollouts re-filtered by an LLM judge) as a stand-in for the on-policy corrections a static dump forbids.

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